FDE ACADEMY · LESSON 2 OF 9

Enterprise RAG

The Credit Policy Intelligence Service: retrieval that knows which version of the rule applies.

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git checkout lesson-02

OBJECTIVES

By the end of today you can

01

Parse eCFR XML, HTML and Federal Register PDFs into labelled sections

02

Chunk without crossing sections, and cite every chunk precisely

03

Combine keyword and vector search with reciprocal rank fusion

04

Filter before ranking, so withdrawn or superseded text is never cited

05

Make the retriever say “no applicable policy found”

06

Measure retrieval and set thresholds from data

THE CORPUS

The lender’s policy plus ten real federal documents

Document	Format	Status	Chunks
MDL internal lending policy (fictional)	Markdown	current	9
Regulation B, 12 CFR 1002 (ECOA)	eCFR XML	current	468
Reg Z §1026.43 ability to repay	eCFR XML	current	43
Reg Z §1026.43 as of June 2020	eCFR XML	superseded	33
FTC Red Flags Rule · FinCEN CIP rule	eCFR HTML	current	31 · 12
CFPB Circulars 2022-03 and 2023-03	FR PDF	withdrawn	12 · 18
CFPB withdrawal notice, May 2025	FR PDF	current	15
Fed SR 26-2 model risk guidance	PDF	current	2 · 21

664 chunks · built in memory in about 2 seconds · no database, no API key

THREE REAL TRAPS

The best match is not always the rule in force

43%

A superseded limit

The 2020 text of Reg Z §1026.43(e)(2)(vi) capped DTI at 43 percent. Since 2021 the same paragraph is a price test.

2025

Withdrawn AI guidance

The two CFPB circulars on explaining algorithmic denials were withdrawn on 12 May 2025. They are still the best keyword match.

2×

A duplicated document

The SR 26-2 letter PDF repeats its whole attachment. Index both and every model-risk answer appears twice.

PARSING · RAG/PARSERS.PY

Each format loses structure differently

eCFR XML

Sections and appendices are tagged, but each paragraph carries only its own level: **(4)**, not **(c)(4)**. The parser rebuilds the full path.

eCFR HTML

Same content; structure lives in CSS classes and data attributes. The full label is in **data-title**.

Federal Register PDF

Three columns, hyphenated line breaks, and other agencies' rules on the same page. Cut at our own **[FR Doc. ...]** line.

```
(a) Notification of action taken…—(1) When notification is required. → label (a)(1)
```

CHUNKING · RAG/CHUNKING.PY

A chunk cites exactly one place

1. Never cross a section boundary

2. Prefer to break at a top-level paragraph, so (b)(2) stays with its sub-paragraphs

3. Cap at 220 words; split long paragraphs at sentence ends

IDS AND CITATIONS

```
reg-b-12cfr1002#1002.9(b)(2)
→ 12 CFR 1002.9(b)(2)

mdl-upl-policy-3.2#4
→ MDL policy §4 Debt-to-income

cfpb-circular-2022-03#Analysis
→ CFPB Circular 2022-03, Analysis
```

HYBRID SEARCH · RAG/RETRIEVER.PY

Two rankings, fused by rank



BM25

Exact terms and citations: “1002.9”, “adverse action”. Written out in 40 lines.

Hashing embedder

Deterministic and offline. Catches spelling overlap, not meaning: a real model plugs into the same port.

Why ranks

BM25 scores and cosines live on different scales. Ranks don’t need calibration.

VERSIONING · IN_FORCE()

Filter first, then rank

A document is in force on date **T** if:

```
in_force_from ≤ T
and T < withdrawn_on
and no successor is in force on T
```

Dates live in the manifest, not the model. The default date is today; **--as-of** answers “what applied when this loan was made?”

Post-filter

Rank everything, drop stale hits. A withdrawn circular takes a top-5 slot, then vanishes: you get 4 results, or none.

Pre-filter (ours)

Stale documents are never candidates. The query about AI denials falls back to Regulation B itself.

ABSTAIN

“No applicable policy found” is an answer

Abstain if either signal is weak:

```
keyword coverage < 0.14
or best cosine < 0.17
```

Coverage counts words the corpus has never seen against the query, so “pasta recipe” looks uncovered, not covered by “fresh”.

Group (eval)	Coverage	Cosine
17 answerable, lowest	0.19	0.25
3 off-topic, highest	0.09	0.10
Threshold (midpoint)	0.14	0.17

MEASURING · UNDERWRITER EVAL-RETRIEVAL

Twenty hand-written cases, four numbers

1.00
hit@5 · a correct source in the top 5

0.89
MRR · how high it ranks

1.00
abstain accuracy

0
stale citations

The weak spot: the 2020 DTI paragraph ranks 5th for its own question. Keywords and hashed vectors both see “debt”, “income” and “ratio” everywhere.

LIVE WALKTHROUGH

Six searches that tell the story

```
uv run underwriter ingest
uv run underwriter search "Applicant DTI = 46%. What policy applies?"
uv run underwriter search "maximum DTI for a qualified mortgage"
uv run underwriter search "…qualified mortgage" --as-of 2020-06-01
uv run underwriter search "credit denials by complex algorithms"
uv run underwriter search "…complex algorithms" --include-withdrawn
uv run underwriter search "best pasta recipe"
uv run underwriter run data/applications/APP-002.json
```

The last command shows the workflow now citing **mdl-upl-policy-3.2#4**. The decision itself still comes from the engine.

LAB · 100 MINUTES

Three exercises

30 MIN · CODE

Write rrf_merge

Empty the function, then make tests/test_rag_fusion.py pass from the docstring. Explain why we fuse ranks, not scores.

40 MIN · EVAL

Grow the eval

Add five cases you verified by reading the source: three answerable, one as-of, one must-abstain. Defend your thresholds.

30 MIN · EXPERIMENT

Chunk size

Run the eval at 120, 220 and 400 words. Record hit@5 and MRR. Which would you ship, and what would you watch?

HOMEWORK AND INTERVIEW PREP

What strong work looks like

Criterion	Strong	Weak
Eval cases	Found by reading the document	Copied from output
Versioning	Explains filter-then-rank	“We drop old docs”
Abstain	Threshold from both groups	A round number
Citations	Paragraph a reviewer can open	Document only

Interview questions

- Design RAG for regulation. How do you handle versions?
- Pre-filter vs post-filter: when does each fail?
- When does BM25 beat embeddings?
- How do you know retrieval is good enough?
- How do you stop RAG answering with nothing relevant?

TAKEAWAY

Retrieval explains and cites. The engine still decides.

Next · Lesson 3

Prompt and context engineering

An LLM writes the recommendation from these citations, in a schema we validate, with a mock model so it runs offline.